



Urban Vertical Aquaponics as a Food Security and Poverty Reduction Strategy

Presented by:

Group-13

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Slide 1 | Urban Vertical Aquaponics Strategy

The Urban Challenges: Pressure on Developing Cities



Rapid Urbanization

Straining resources, shrinking viable land.



Food Insecurity & Rising Prices

Millions lack access to safe, nutritious food.



Growing Unemployment

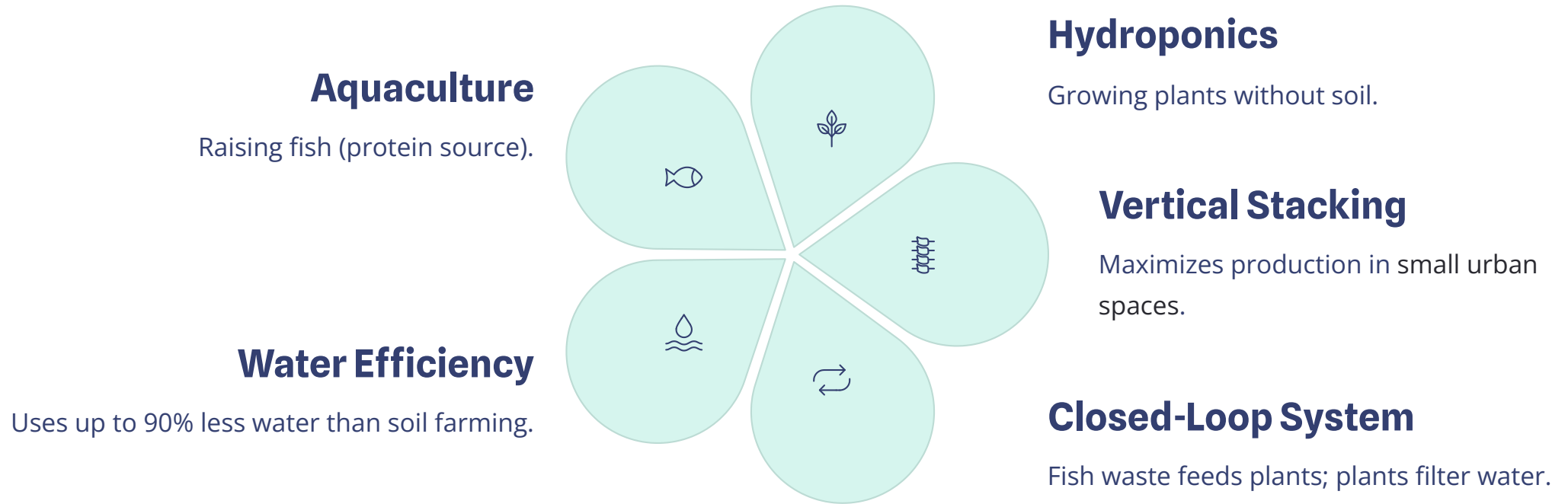
Limited job opportunities in dense urban centers.



Unsustainable Traditional Farming

Cannot meet the pace of urban demand and expansion.

Concept: Urban Vertical Aquaponics (UVA)



Conceptual Model of the System

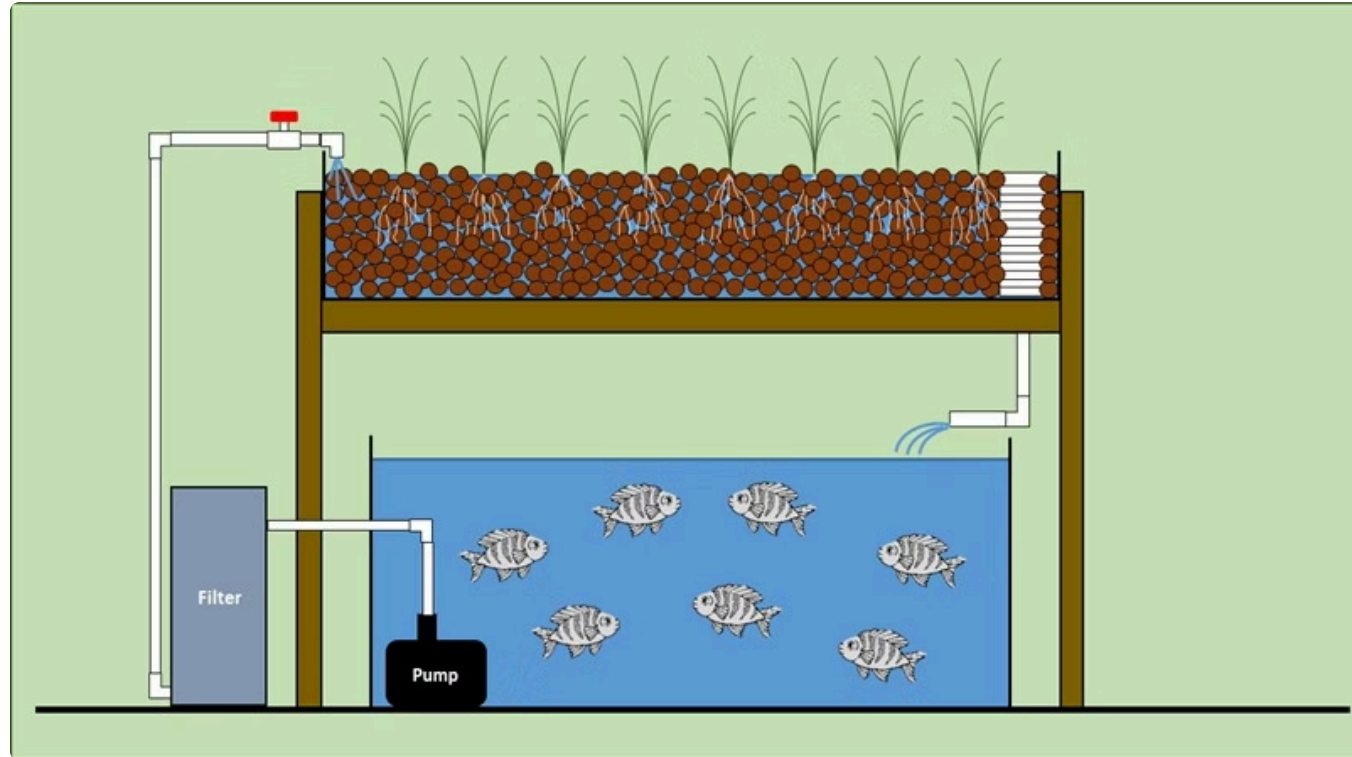


Figure 1: Conceptual model of Urban Vertical Aquaponics system

Strengths and Opportunities: Efficiency and Impact

Strengths

Space Utilization

High-density farming in limited area.

Cost & Carbon Reduction

Minimal transport; lower environmental footprint.

Local Job Creation

Training and operation roles for community.

Opportunities

Economic Viability

Meeting rising demand for local, organic food.

Energy Independence

Integration with solar or biogas renewable energy.

Community Integration

Educational platforms and micro-enterprise.

Challenges and Threats: Barriers to Adoption

1

High Initial Investment

Cost of systems, pumps, and infrastructure.

2

Technical Dependency

Requires specialized skills for maintenance and monitoring.

3

Policy & Regulation Gaps

Lack of support for funding and training access.

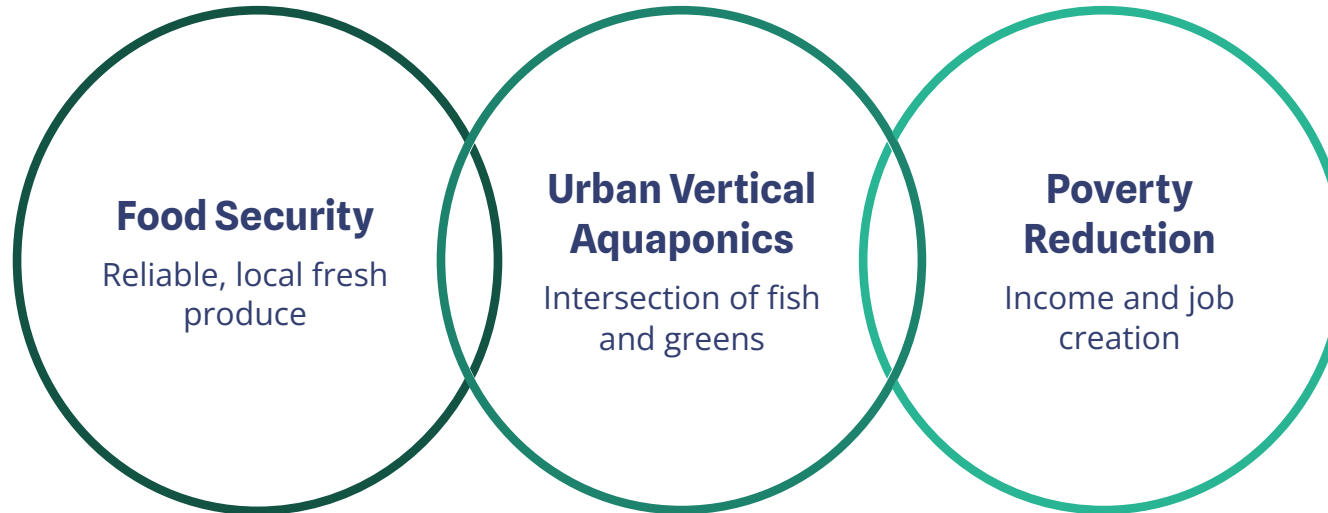
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Environmental Risks

Vulnerability to power failures and extreme weather (heatwaves).

Conclusion: A Path to Urban Resilience

Urban Vertical Aquaponics is a sustainable solution that offers dual benefits:



Key Takeaways

- Simple technology, profound environmental and social benefits.
- Transforms underutilized urban spaces into productive assets.
- Requires supportive policy, education, and public awareness for success.

Thank You



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